b. Lower bound on curvature

Now the curvature can be bounded using strong convexity arguments as in Lemma 4. For this we first connect δH to δf, with f as defined in (F28),

$$\delta H(\Delta_u, \theta_u^*) = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu}\left(f(\sigma_u(\langle\hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u}\rangle + \theta_u)) - f(\sigma_u(\langle\hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u}\rangle + \theta_u^*)) - \Delta_u\sigma_u f'(\sigma_u(\langle\hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u}\rangle + \theta_u^*))\right), \tag{F62}$$

$$= \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \delta f(\Delta_u\sigma_u, \theta_u^*). \tag{F63}$$

Hence from the lower bound in Lemma 4,

$$\delta H(\Delta_u, \theta_u^*) \geq \frac{e^{-2h_{max}}}{2} \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} (\Delta_u\sigma_u)^2 = \frac{e^{-2h_{max}}}{2}(\Delta_u)^2 \tag{F64}$$

Now from the upper bound in Lemma 4, we have $0 < \delta H(\Delta_u, \theta_u^*) \leq \frac{h_{max}^2}{2}$.

We can use these these bounds with Hoeffding's inequality (Lemma 8) to get the following probabilistic upper-bound on the estimated curvature,

$$Pr\left(\delta H_M(\Delta_u, \theta_u^*) \leq \delta H(\Delta_u, \theta_u^*) - \epsilon_b\right) \leq e^{\frac{-8\epsilon_b^2 M}{h_{max}^4}}. \tag{F65}$$

Hence ,$M \geq \frac{h_{max}^4}{8\epsilon_b^2}\log(\frac{2}{\delta})$ ensures that the corresponding lower bound holds w.p greater than $1 - \frac{\delta}{2}$.

c. Magnetic field learning guarantee

Now using the bounds on gradient and curvature in (F53) and using the union bound, we have the following bound that holds with probability $1 - \delta$ when M is greater than $\max\left(\frac{h_{max}^4}{8\epsilon_b^2}\log(\frac{2}{\delta}), \frac{2}{\epsilon_a^2}\log(\frac{4}{\delta})\right)$,

$$\frac{e^{-2h_{max}}}{2}\Delta_u^2 \leq \epsilon_b + 2h_{max}\left(4d\varepsilon + \frac{4\eta}{\omega_P} + \epsilon_a\right) \tag{F66}$$

Now choosing $\frac{\varepsilon_h^2}{2} = 2e^{2h_{max}}\epsilon_b = 4h_{max}e^{2h_{max}}\epsilon_a$, gives us

$$\Delta_u \leq \varepsilon_h + 4\sqrt{d\varepsilon h_{max}}e^{h_{max}} + 4\sqrt{\frac{\eta h_{max}}{\omega_P}}e^{h_{max}}. \tag{F67}$$

Plugging the definition of ε_h in the M values, we see that this can be achieved with $M = \left\lceil \frac{2^7 h_{max}^4 e^{4h_{max}}}{\varepsilon_h^4} \right\rceil$ samples.